Lecture 07 - Introduction

Lecture 07 - Confidence Intervals and Statistical Precision - Introduction

Purpose of the Lesson

This lecture introduces sampling distributions and confidence intervals. The goal is to move from describing data to making careful statements about a larger population or process when only a sample is observed.

The motivating problem is simple. We rarely observe everything. A manufacturer cannot destroy every rope to test its strength. A company cannot survey every customer. A financial analyst cannot observe all future returns. Instead, we use samples. The sample gives evidence, but the estimate computed from that sample is not perfectly certain.

The lecture therefore introduces two connected ideas. First, different samples from the same population can produce different estimates. This variation is called sampling fluctuation, and it is described by a sampling distribution. Second, confidence intervals use the sampling distribution to report the precision of an estimate.

Python is used later as a practical support tool for computing intervals from real datasets. The conceptual goal is to understand what the interval measures, how standard error enters the calculation, and what careful interpretation sounds like.

Learning Objectives

This lecture aims to:

1. Explain why sampling is necessary in statistical inference.
2. Distinguish sampling from inference.
3. Compare random, systematic, and quota sampling.
4. Explain sampling fluctuation and the sampling distribution of the sample mean.
5. Distinguish standard deviation from standard error.
6. Use the formula (SE(\bar X)=\sigma/\sqrt n) and its estimated version (s/\sqrt n).
7. Explain why larger samples produce more precise estimates.
8. Introduce the Central Limit Theorem as a reason normal approximations are useful for sample means.
9. Construct simple confidence intervals for means.
10. Interpret confidence intervals carefully and avoid common mistakes.

Learning Outcomes

By the end of this session, students should be able to:

1. Identify the population, sample, parameter, estimator, and estimate in an inference problem.
2. Explain why different random samples produce different sample means.
3. Compute a standard error for a sample mean.
4. Explain the difference between individual variability and estimation uncertainty.
5. Use normal critical values for 90%, 95%, and 99% central intervals.
6. Construct a confidence interval using estimate plus or minus critical value times standard error.
7. Explain why a higher confidence level gives a wider interval.
8. Explain why increasing sample size narrows the interval.
9. Interpret a confidence interval as uncertainty around a population mean, not a range for individual observations.
10. Write a cautious conclusion when an interval includes an important benchmark.

Lesson Roadmap

The lecture begins with a concrete sampling problem: testing rope strength without destroying all ropes. This example introduces the distinction between a sample and the population we care about.

The next step is sampling design. We consider random sampling, systematic sampling, and quota sampling. The main lesson is that sampling choices affect the quality and generality of the inference.

The lecture then introduces sampling fluctuation. If we take many samples from the same population, each sample gives its own estimate. Those estimates form a sampling distribution. For the sample mean, the spread of that sampling distribution is called the standard error.

After that, we discuss the Central Limit Theorem. Even when raw data are not normally distributed, sample means are often approximately normal when the sample size is large enough. This justifies the normal critical values used in simple confidence intervals.

The final part constructs confidence intervals. We use common central normal cutoffs, especially 1.645 for 90%, 1.96 for 95%, and 2.58 for 99%. The lecture ends with careful interpretation: a confidence interval describes uncertainty around an estimated parameter, not the range of individual future observations.

Central Idea

The central idea of the lecture is:

A point estimate should not travel alone. It should be accompanied by a measure of precision.

For example, saying that the average return is 0.6% is incomplete. If the standard error is 0.1%, the estimate is relatively precise. If the standard error is 0.8%, the estimate is much less precise. The same point estimate can support very different conclusions depending on the uncertainty around it.

This habit is central to professional statistical reporting. It is not enough to compute an average. We must also explain how much that average might vary from sample to sample.

How to Work Through the Lesson

Begin by focusing on the sampling logic. Ask what population we care about, what sample we observe, and what parameter we are trying to estimate.

Then study the difference between standard deviation and standard error. This is one of the most important distinctions in the course. Standard deviation describes dispersion of individual observations. Standard error describes uncertainty in an estimate.

Next, practice the formula (SE=s/\sqrt n). Notice how increasing the sample size reduces the standard error, but only at the square-root rate.

Finally, work through confidence intervals slowly. Identify the estimate, the standard error, the critical value, and the margin of error. Then write a careful interpretation in words.

What to Pay Attention To

Three habits matter in this lecture.

First, check the sampling process. A large sample can still be misleading if it is biased or unrepresentative.

Second, distinguish variability from precision. A dataset can have highly variable observations, but a large sample may still estimate the mean precisely. Conversely, a small sample can give a noisy estimate even if the sample mean looks impressive.

Third, interpret intervals carefully. A confidence interval for a mean is not a prediction interval for individual observations. It is an interval describing uncertainty around a population average.

End-of-Lesson Check

By the end of the lecture, you should be comfortable answering questions such as:

1. What is the difference between sampling and statistical inference?
2. Why can different samples from the same population give different estimates?
3. What is a sampling distribution?
4. What is the difference between standard deviation and standard error?
5. Why does the standard error decrease with sample size?
6. What does the Central Limit Theorem say about sample means?
7. What are the common 90%, 95%, and 99% normal critical values?
8. How do we construct a confidence interval for a mean?
9. Why does a higher confidence level produce a wider interval?
10. Why is it incorrect to say that a confidence interval for a mean contains 95% of individual observations?

If these questions are clear, you are ready to use confidence intervals as the foundation for hypothesis testing and regression inference.